Pragathi Praveena

Human-Computer and Human-Robot Interaction Researcher

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RESEARCH OVERVIEW

I am interested in developing novel interactive systems, particularly **collaborative systems** that leverage advancements in robotics and AI to facilitate human-human interactions. My work draws on methods and concepts from Human-Computer Interaction (HCI), Human-Robot Interaction (HRI), and Computer-Supported Cooperative Work (CSCW) to develop complex interactive systems that embody my research ideas and enable empirical investigations. Additionally, I am interested in the creation of **enabling resources**, such as datasets, tools, libraries, and languages, to support the development of future interactive systems.

CURRENT POSITION

2024 — Present **Postdoctoral Fellow**, Robotics Institute, Carnegie Mellon University
PI: Reid Simmons

EDUCATION

2017 — 2024 **M.S. and Ph.D. in Computer Sciences**, University of Wisconsin–Madison, USA
Dissertation Title: Towards Effective Robotic Groupware
Committee: Bilge Mutlu (co-chair), Michael Gleicher (co-chair), Michael Zinn, Robert Radwin

2011 — 2015 **Bachelor of Technology in Electrical Engineering**, Indian Institute of Technology Madras, India

Grants

2023 **Google Award for Inclusion Research**, Google Inc.
Co-authored research proposal with Bilge Mutlu (PI), **$60,000**
Topic: Supporting Social Participation for Older Adults through Robotic Telepresence

2023 **Expanding Our Vision Award**, McPherson Eye Research Institute, UW–Madison
Co-authored research proposal with Bilge Mutlu (PI), **$10,000**
Topic: Designing Interfaces to Enhance the Experience of Remote Vision through Robotic Cameras

Honors & Awards

2024 **Rising Stars in EECS**, Massachusetts Institute of Technology *(19% acceptance)*

2024 **Special Recognition for Outstanding Review**, ACM Conference on Designing Interactive Systems (DIS)

2023 **ACM SIGCHI Gary Marsden Travel Award**
Selective award for full support to attend ACM Conference on Human Factors in Computing Systems (CHI)

2023 **HRI Pioneer**, ACM/IEEE Conference on Human-Robot Interaction (HRI)
Fully funded participant in selective doctoral consortium *(25% acceptance)*

2020 **RSS Pioneer**, Robotics: Science and Systems (RSS)
Fully funded participant in selective doctoral consortium *(32% acceptance)*

2020 **Best Paper Award Finalist** *(top 5%)*, ACM/IEEE Conference on Human-Robot Interaction (HRI)

2016 **Xerox Patent Award**, Awarded by Xerox to the lead inventor on a filed patent

2015 **Institute Blues** *(top 3 in ~800 graduates)*, IIT Madras
Motorola Prize *(#1 in ~150 EE and CS graduates),* IIT Madras
Recognized for exceptional overall achievement during undergraduate studies

2014 **French Government Charpak Scholarship**
Two months of support for research experience at École Normale Supérieure, Paris

WORK & RESEARCH EXPERIENCE

2024 — Present **Postdoctoral Fellow**, Robotics Institute, **Carnegie Mellon University**
NSF AI Institute for Collaborative Assistance and Responsive Interaction for Networked Groups

2017 — 2024 **Graduate Researcher**, People and Robots Lab, **University of Wisconsin–Madison**
Designed, built, and evaluated human-robot interfaces to enable remote and collaborative work
Expanded a grant's original scope by initiating a new research direction that combined HRI & CSCW

2015 — 2017 **Junior Research Scientist**, Data Analytics Lab, **Xerox Research Centre India**
Developed and evaluated novel algorithms to estimate respiratory patterns using a webcam
Patents licensed by a California-based baby monitor startup

Spring 2015 **Undergraduate Researcher**, Assistive Technology Lab, **Indian Institute of Technology Madras**

Summer 2014 **Undergraduate Researcher**, Group for Neural Theory, **École Normale Supérieure, France**

Summer 2013 **Project Intern**, Electrical and Electronics Maintenance, **Bosch India**

PUBLICATIONS

* *indicates equal contribution* <u>*indicates students I mentored*</u>

JOURNAL ARTICLES

[J4] IEEE Access '24 <u>Wang, Y.</u>, **Praveena, P.**, & Gleicher, M. "A Design Space of Control Coordinate Systems in Telemanipulation." *IEEE Access.*

[J3] CSCW '23 **Praveena, P.**, Wang, Y., Senft, E., Gleicher, M., & Mutlu, B. "Periscope: A Robotic Camera System to Support Remote Physical Collaboration." *Proceedings of the ACM on Human-Computer Interaction, 7*(CSCW2).

[J2] Human Factors '22 Ramesh, B., Konstant, A., **Praveena, P.**, Senft, E., Gleicher, M., Mutlu, B., Zinn, M., & Radwin, R.G. "Manually Acquiring Targets from Multiple Viewpoints Using Video Feedback." *Human Factors.*

[J1] TSP '17 Prathosh, A.P., **Praveena, P.**, Mestha, L.K., & Bharadwaj, S. "Estimation of Respiratory Pattern from Video Using Selective Ensemble Aggregation." *IEEE Transactions on Signal Processing.*

REFEREED FULL CONFERENCE PAPERS

[C11] DIS '24 <u>Lee, C. P.</u>, **Praveena, P.**, & Mutlu, B. "Designing User-centered Repair and Explanations to Address Robot Failures." *ACM Conference on Designing Interactive Systems.*

[C10] ICRA '23 <u>Wang, Y.</u>, **Praveena, P.**, Rakita, D., & Gleicher, M. "RangedIK: An Optimization-Based Robot Motion Generation Method for Ranged-Goal Tasks." *IEEE International Conference on Robotics and Automation.*

[C9] IROS '22 Senft, E.*, Hagenow, M.*, **Praveena, P.**, Radwin, R., Zinn, M., Gleicher, M., & Mutlu, B. "A Method for Automated Drone Viewpoints to Support Remote Robot Manipulation." *IEEE/RSJ International Conference on Intelligent Robots and Systems.*

[C8] HRI '22 **Praveena, P.**, Molina, L., Wang, Y., Senft, E., Mutlu, B., & Gleicher, M. "Understanding Control Frames in Multi-Camera Robot Telemanipulation." *ACM/IEEE International Conference on Human-Robot Interaction.*

[C7] HRI '20 **Praveena, P.**, Rakita, D., Mutlu, B., & Gleicher, M. "Supporting Perception of Weight through Motion-induced Sensory Conflicts in Robot Teleoperation." *ACM/IEEE International Conference on Human-Robot Interaction.* 🎖 [Best Paper Award Finalist]

[C6] ICRA '19 **Praveena, P.**, Rakita, D., Mutlu, B., & Gleicher, M. "User-Guided Offline Synthesis of Robot Arm Motion from 6-DoF Paths." *IEEE International Conference on Robotics and Automation.*

[C5] HRI '19 **Praveena, P.**, Subramani, G., Mutlu, B., & Gleicher, M. "Characterization of Input Methods for Human-to-robot Demonstrations." *ACM/IEEE International Conference on Human-Robot Interaction.*

[C4] BIBE '16 Chatterjee, A., Prathosh, A.P., **Praveena, P.**, & Upadhya, V. "Real-time Visual Respiration Rate Estimation with Dynamic Scene Adaptation." *IEEE International Conference on Bioinformatics and Bioengineering.*

[C3] BIBE '16 Chatterjee, A., Prathosh, A.P., **Praveena, P.**, & Upadhya, V. "A Vision Based Method for Real-time Respiration Rate Estimation Using a Recursive Fourier Analysis." *IEEE International Conference on Bioinformatics and Bioengineering.*

[C2] BIBE '16 Upadhya, V., Chatterjee, A., Prathosh, A.P., & **Praveena, P.** "Respiration Monitoring through Thoraco-Abdominal Video with an LSTM." *IEEE International Conference on Bioinformatics and Bioengineering.*

[C1] EMBC '16 Chatterjee, A., Prathosh, A.P., & **Praveena, P.** "Real-time Respiration Rate Measurement from Thoracoabdominal Movement with a Consumer Grade Camera." *IEEE International Conference of the Engineering in Medicine and Biology Society.*

Juried short conference papers/Workshop papers/Extended abstracts

[S6] HRI '24 Hwang, Y., Sato, A. J., **Praveena, P.**, & Mutlu, B. "Understanding Generative AI in Robot Logic Parametrization." *Workshop at ACM/IEEE International Conference on Human-Robot Interaction on End-User Development for Human-Robot Interaction.*

[S5] AAAI FSS '23 **Praveena, P.**, Schoen, A., Gleicher, M., Porfirio, D., & Mutlu, B. "Petri Nets for the Iterative Development of Interactive Robotic Systems." *AAAI Fall Symposium on Unifying Representations for Robot Application Development.*

[S4] CSCW '23 Meng, H., Wang, Y., **Praveena, P.**, Gleicher, M., & Mutlu, B. "Demonstrating Periscope: A Robotic Camera System to Support Remote Physical Collaboration." *Demonstration at ACM Conference On Computer-Supported Cooperative Work and Social Computing.*

[S3] CHI '23 **Praveena, P.***, Cagiltay, B.*, Gleicher, M., & Mutlu, B. "Exploring the Use of Collaborative Robots in Cinematography." *Late-Breaking Work at ACM Conference on Human Factors in Computing Systems.* 🎖 [ACM SIGCHI Gary Marsden Travel Award]

[S2] HRI '23 **Praveena, P.**, Gleicher, M., & Mutlu, B. "Designing Robotic Camera Systems to Enable Synchronous Remote Collaboration." *Extended Abstract at ACM/IEEE International Conference on Human-Robot Interaction.* 🎖 [HRI Pioneer]

[S1] RSS '20 **Praveena, P.**, Mutlu, B., & Gleicher, M. "Human-Robot Interfaces for Physical Interactions." *Extended Abstract at Robotics: Science and Systems.* 🎖 [RSS Pioneer]

Other publications

[M2] HCIC '18 **Praveena, P.**, Mutlu, B., & Gleicher, M. "Communicating Physical Interactions to Robots." *Human–Computer Interaction Consortium Workshop: AI and HCI.*

[M1] BEATS '14 **Praveena, P.**, Kavalam, J., & Jacob, N. "A smartphone-based vision simulator." *International Conference on Biomedical Engineering and Assistive Technologies.*

Patents

[P2] "System and method for extracting a periodic signal from video." 2019. US Patent 10,192,307.
[P1] "Determining respiration rate from a video of a subject breathing." 2018. US Patent 9,861,302.

MENTORING

PhD Students

People and Robots Lab, UW–Madison (Advised jointly w/ Bilge Mutlu)

2023 — Present **Yaxin Hu**, Conversational mobile robots for remote exploration
2023 — Present **Nathan White**, AI-assisted qualitative analysis of multimodal data
2023 — Present **Dakota Sullivan**, Primal world beliefs for robots
2023 — Present **Yuna Hwang**, Expressive robot motion generation using natural language; **Author:** S6
2023 — Present **Christine Lee**, Explanations and repair for unexpected robot behavior; **Author:** C11

Graphics Lab, UW–Madison (Advised jointly w/ Michael Gleicher)

2022 — 2023 **Yeping Wang**, Robot motion generation for ranged-goal tasks; **Author:** C10, J4

Research staff, UW–Madison

2020 — 2021 **Luis Molina**, Multi-camera interface for robot telemanipulation; **Author:** C8

Undergraduate students

People and Robots Lab, UW–Madison (Advised solely or jointly w/ graduate student)

2024 — Present **Nikhil Kruthiventi**, *(w/ Nathan White)* AI assistance for qualitative analysis
2024 — Present **Taenam Kim**, *(w/ Nathan White)* User experience for AI-assisted qualitative analysis
2023 — Present **Rainy Jin**, User experience for specifying system logic using Petri nets
2023 — Present **Zejun Zhou**, Collaborative editor for specifying system logic using Petri nets
2023 — Present **Sydney Scalzo**, Embodied behavior as a creative medium
2022 — 2023 **Haoming Meng**, *(w/ Yeping Wang)* Web interface for robot-supported collaboration; **Author:** S4
2022 **Lily Reback**, Qualitative coding of multimodal data; **Acknowledgement:** J3

Graphics Lab, UW–Madison (Advised jointly w/ Michael Gleicher)

2022 **Alexander Peseckis**, Web-based robot motion viewer
2022 **William Cong**, VR platform for robot experiments
2021 — 2022 **Gia-phong Nguyen**, Web-based robot motion viewer
2021 — 2022 **Sage Livingstone**, Blender plugin for creating models of robots and scenes
2019 — 2020 **Jack Yang**, VR platform for robot experiments
2019 — 2020 **Sayem Wani**, Web interface for MOVO, a mobile bi-manual robot
2019 — 2020 **Joshua Mathews**, Control system for MOVO, a mobile bi-manual robot

Peer mentoring

People and Robots Lab, UW–Madison

April 2022 — August 2023

I organized a peer mentorship program in which 2–3 graduate students met with a different student mentor each week. Through this program, I provided peer mentorship to 12 graduate students through **weekly sessions**.

Mentees: Yuna Hwang, Hailey Johnson, Amy Koike, Callie Kim, Christine Lee, Dakota Sullivan, Irene Ho, Bengisu Cagiltay, Yaxin Hu, Nathan White, Nitzan Orr, Kevin Welsh

Acknowledgments: Original concept by Bengisu Cagiltay, currently organized by Hailey Johnson (from Aug 2023)

Teaching experience

Summers 2018, 2019 — **Social Robotics Instructor,** Grandparents University, UW–Madison
Co-designed and co-taught lecture and lab session for children and their grandparents

Fall 2017 — **Teaching Assistant**, ECE 203: Signals, Information and Computation, UW–Madison
Co-taught in a flipped classroom, held office hours, managed online Q&A for ~200 students

2014 — 2015 — **President and Instructor**, Web Operations Club, Centre for Innovation, IIT Madras
Co-organized introductory and advanced sessions for 400+ students on web development
Taught graphics editing and design thinking to 100+ students for web & app development

Invited talks

October 2023 — Adaptive Systems Section, **Naval Research Laboratory**, Washington, DC, USA | *Host:* Laura Hiatt
"Supporting Remote Human Collaboration for Physical Tasks through Robotic Cameras"

October 2023 — Intuitive Computing Lab, **Johns Hopkins University**, Baltimore, MD, USA | *Host:* Chien-Ming Huang
"Supporting Remote Human Collaboration for Physical Tasks through Robotic Cameras"

April 2023 — CS Departmental Research Symposium, **UW–Madison**, Madison, WI, USA 🏅 [Best Talk Award]
"Designing Robotic Camera Systems to Enable Synchronous Remote Collaboration"

April 2021 — LUCID Seminar, **UW–Madison**, Madison, WI, USA [Virtual]
"Visual Awareness in Remote Environments"

Academic service

Program committee

2025 — Computer-Supported Cooperative Work & Social Computing (CSCW)
2024 — Pioneers Workshop at ACM/IEEE International Conference on Human-Robot Interaction (HRI)

Referee service

2021 — 2024 — ACM/IEEE International Conference on Human-Robot Interaction (HRI)
2023 — 2024 — ACM Conference on Human Factors in Computing Systems (CHI)
2024 — ACM Conference on Designing Interactive Systems (DIS) 🏅 [Special Recognition]
2023 — Computer-Supported Cooperative Work & Social Computing (CSCW)

Ad-hoc Reviewer

2024 — IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
2023 — AAAI Fall Symposium on Unifying Representations for Robot Application Development
2023 — Automation in Construction
2022 — IEEE Robotics and Automation Letters (RA-L)

Organization

2024 **Co-chair** (w/ Nina Moorman), AAAI Fall Symposium on AI for Aging in Place, Arlington, VA, USA
Organizing a two and one-half day multidisciplinary symposium

2024 **Organizer**, HRI and UIST: Designing Socially Engaging Robot Interfaces, Pittsburgh, PA, USA
Leading an interactive workshop centered on social robotics

2024 **Panel/Networking Chair**, Pioneers Workshop at HRI, Boulder, CO, USA
Initiated the HRI Pioneers Mentorship Program

2021 **Local/Social Chair**, Pioneers Workshop at RSS, Virtual

Outreach

2023 **Staff**, UW–Madison CS recruitment booth, Grace Hopper Celebration, Orlando, FL, USA

Summer 2022 **Organizer** (along with Yaxin Hu), Human-Centered Computing Reading Group, UW–Madison

2018 — 2023 **Volunteer**, Lab tours & demos for visiting school children, graduate students, & faculty candidates

Extra-curricular

2019 **Morgridge Entrepreneurial Bootcamp**, UW–Madison
Selected to attend a one-week training program in technology entrepreneurship for graduate students

2018 **gALPHA Entrepreneurship Program**, UW–Madison
Selected to attend a four-week venture-creation program by *gener8tor*, a nationally ranked accelerator

2018 **Hackathon winner** *(#1 in 8 teams)*, EnerHack, UW–Madison

2014 **Hackathon winner** *(#1 in ~20 teams)*, Geek Up, IIT Madras; Invited to present at Google DevFest, Chennai